

Research Memo: Assessing Outlier Influence on Communication Apprehension and Public Transit Usage

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Research or Analytical Question

Are there any statistical outliers present in the data, and to what extent do these potential outliers exert leverage or influence on the fitted regression line between communication apprehension and public transit usage?

Answer, Response, + Summary of Results

To determine whether extreme observations distort the analysis, we evaluated bivariate regression influence using Cook’s Distance (threshold $4/N = 0.0159$) across $N = 252$ complete observations.

Table 1: Comparison of OLS regression models before and after outlier removal

	Model	Intercept	Slope (CA Score)	p-value	R-squared
0	Full Model (N=252)	6.982	-0.986	0.001	0.043
1	Pruned Model (N=245)	7.849	-1.459	0.000	0.094

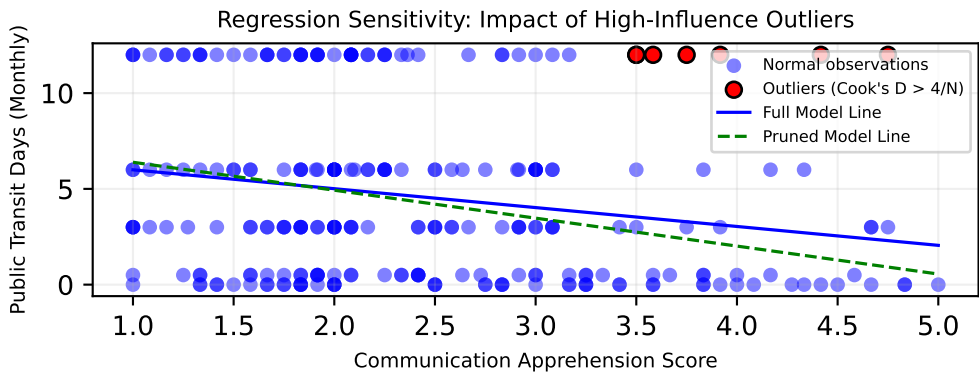


Figure 1: Scatter plot with regression lines before and after outlier removal

Summary of Results

- **Key Finding:** As shown in Table 1, we identified **7 influential observation(s)** ($4/N = 0.0159$). Removing them strengthened the negative slope from **-0.986** ($p = 0.001$) to **-1.459** ($p < 0.001$), with R^2 moving from **0.043** to **0.094**.
- **Conclusion:** As illustrated in Figure 1, removing high-influence outliers does not alter the core finding: communication apprehension has a **statistically significant negative relationship** with transit usage. Removing outliers reveals a stronger underlying negative effect.

What questions or uncertainties remain?

1. **Measurement Limitations:** Converting ordinal category choices into exact monthly days via midpoints creates discrete cluster bands. Would **Logistic Regression** (predicting binary regular usage) better accommodate these extreme boundary values?
2. **Omitted Variable Bias:** The identified outliers may reflect unmeasured socio-demographic factors (e.g., vehicle ownership, transit access, income) rather than true anomalies in communication apprehension.